

Annah Singh

EE 472

11/11/24

Homework #6

Task

✓

▼

▶

Compile Design

✓

>

▶

Analysis & Synthesis

✓

>

▶

Fitter (Place & Route)

✓

>

▶

Assembler (Generate programming file)

✓

>

▶

Timing Analysis

>

▶

EDA Netlist Writer

■

Edit Settings

▶

Program Device (Open Programmer)

```
1 module hw6 (output reg [4:0] count_out,  
2             input wire rst_n, preset, clk, up_down,  
3             input wire [4:0] count_in);  
4     always @ (posedge clk or negedge rst_n or posedge preset)  
5     begin  
6         if (!rst_n)  
7         begin  
8             count_out <= 4'b0000;  
9         end  
10        else if (preset)  
11        begin  
12            count_out <= count_in;  
13        end  
14        else if (up_down)  
15        begin  
16            if (count_out % 2'd2 == 1'b0)  
17            begin  
18                count_out <= count_out + 1'd1;  
19            end  
20            else  
21            begin  
22                count_out <= count_out + 2'd2;  
23            end  
24        end  
25        else if (!up_down)  
26        begin  
27            if (count_out % 2'd2 == 1'b0)  
28            begin  
29                count_out <= count_out - 2'd2;  
30            end  
31            else  
32            begin  
33                count_out <= count_out - 1'd1;  
34            end  
35        end  
36    end  
37 endmodule
```



